

Framework structure

Layer 1: Benchmark Input Layer

This is the foundation of the framework.

Purpose

To ensure all systems are evaluated on the same high-quality iTaukei–English benchmark set.

Inputs

- source sentence
- reference translation
- system output
- domain
- sentence type
- direction

Recommended metadata

- id
- domain
- sentence_type
- direction
- source_lang
- target_lang
- source
- reference
- hypothesis
- system

Justification:

Without a controlled benchmark set, evaluation becomes inconsistent and difficult to compare.

Layer 2: Automatic Evaluation Layer

This layer provides scalable quantitative assessment.

Metrics to include

- **BLEU** — n-gram overlap
- **chrF++ / chrF** — character-level similarity
- **TER** — edit distance / correction burden
- **COMET** — semantic quality
- **BERTScore** — contextual semantic overlap

Purpose

To capture different aspects of translation quality:

- lexical closeness
- structural similarity
- semantic adequacy

Output

For each translation:

- BLEU
- chrF
- TER
- COMET
- BERTScore

Strength

Fast, replicable, comparable across models

Limitation

May miss cultural nuance and contextual appropriateness

Layer 3: Human Evaluation Layer

This is essential for iTaukei because cultural and contextual meaning matters.

Recommended dimensions

Usage of 1–5 scale for each:

1. Fluency

Is the translation grammatically natural and readable?

2. Adequacy

Does it preserve the meaning of the source?

3. Cultural Appropriateness

Does it reflect culturally acceptable and contextually suitable iTaukei usage?

4. Semantic Fidelity

Does it preserve intended meaning without distortion?

Recommended evaluators

- bilingual iTaukei–English speakers
- ideally 2 or more annotators

Reliability

- Cohen's Kappa or weighted Kappa
- inter-annotator agreement

Output

- mean score per dimension
 - total human score
 - agreement statistics
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Layer 4: LLM-Based Evaluation Layer

- GPT-based evaluator generates:
 - Quality score
 - Explanation (rationale)
 - Prompt structure:
 - Source + reference + hypothesis
 - Scoring criteria aligned with human evaluation
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Layer 5: Error Analysis Layer

Purpose

To explain *why* systems fail, not just *how much* they fail.

Proposed error taxonomy

A. Lexical errors

- wrong word choice
- untranslated word
- incorrect terminology

B. Syntactic errors

- word order problems
- clause structure issues
- agreement errors

C. Semantic errors

- meaning shift
- omission
- addition
- mistranslation

D. Pragmatic / discourse errors

- wrong tone
- incorrect contextual meaning
- poor cohesion

E. Cultural errors

- inappropriate idiom rendering
- culturally unnatural phrasing
- failure to preserve culturally embedded meaning

Error annotation format

For each problematic translation:

- error type
- severity
- notes

Severity levels

- minor
- moderate
- major
- critical

Layer 6: Integrated Interpretation Layer

Final decision-making part of the framework.

Purpose

To combine evidence from all previous layers into an interpretable evaluation outcome.

You can produce:

- sentence-level evaluation
- system-level evaluation

- domain-level evaluation
- direction-level evaluation

Possible outputs

- best system overall
- best system by domain
- weakest domain for each system
- most common error types
- correlation between automatic and human scores